

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
ELECTRONICS AND COMMUNICATION ENGINEERING

VII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
22HS1MG201	Engineering Economics and Accountancy	3	0	0	3	3
22PC1EC401	Microwave Engineering	3	0	0	3	3
	PROFESSIONAL ELECTIVE – III					
22PE1EC401	Cellular and Mobile Communications	3	0	0	3	3
22PE1EC402	DSP Processors and Architectures					
22PE1EC403	Languages for Embedded Software					
22PE1EC404	VLSI Scripting Languages					
22PE1EC405	Organic Electronics					
	PROFESSIONAL ELECTIVE – IV THROUGH SWAYAM-NPTEL					
	Communication Domain	3	0	0	3	3
	Signal Processing Domain					
	General Domain					
	VLSI Domain					
	Embedded Systems Domain					
	OPEN ELECTIVE – III	3	0	0	3	3
22PC2EC401	Microwave Engineering Laboratory	0	0	2	2	1
22PC2EC304	Machine Learning laboratory	0	0	2	2	1
22PW4EC401	Major Project Phase – I	0	0	6	6	3
Total		15	0	10	25	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD
B.TECH. IV YEAR
ELECTRONICS AND COMMUNICATION ENGINEERING

VIII SEMESTER

R22

Course Code	Title of the Course	L	T	P/D	CH	C
	PROFESSIONAL ELECTIVE – V					
22PE1EC406	Wireless Communications and Networks	3	0	0	3	3
22PE1EC407	Speech Signal Processing					
22PE1EC408	Radar Systems					
22PE1EC409	Introduction to ASIC Design					
22PE1EC410	Application of Data Analytics in IoT					
	PROFESSIONAL ELECTIVE – VI					
22PE1EC411	Satellite Communications	3	0	0	3	3
22PE1EC412	5G Communication					
22PE1EC413	Electronic Measurements and Instrumentation					
22PE1EC414	Low Power VLSI Systems					
22PE1EC415	IoT and Raspberry Pi Programming					
	OPEN ELECTIVE – IV	3	0	0	3	3
22PW4EC402	Major Project Phase – II	0	0	22	22	11
Total		9	0	22	31	20

L – Lecture T – Tutorial P – Practical D – Drawing
C – Credits SE – Sessional Examination CA – Class Assessment
SEE – Semester End Examination D-D – Day to Day Evaluation
CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
ELA – Experiential Learning Assessment
LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22HS1MG201) ENGINEERING ECONOMICS AND ACCOUNTANCY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To understand the basic concepts of economics and different forms of business organizations
- To create awareness on basics of business economics and to analyze the concepts of demand and supply
- To describe each stage of product life cycle with the help different costs and their role in maintaining optimum cost of production and overall profitability by considering different market competitions
- To acquaint with the basic accounting knowledge and financial accounting process
- To evaluate the performance of the organization using various ratios

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Perform decision making function effectively in an uncertain framework by applying the based concepts of economics and select suitable form of business organization which meets the requirements of business

CO-2: Take the right decisions towards buying and selling of goods and services based on the demand and supply dynamics in the markets

CO-3: Fix the right price based upon production cost which can best meet the pre-determined objectives of the business under different market conditions

CO-4: Prepare book of accounts and understand overall position of the business

CO-5: Interpret the firm's financial performance using various ratios

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	-	-	-	-	-	-	-	1	2	1	3	1	1	1	1
CO-2	-	-	-	-	-	-	-	1	2	1	3	1	1	1	1
CO-3	-	-	-	-	-	-	-	1	2	1	3	1	1	1	1
CO-4	-	-	-	-	-	-	-	1	2	1	3	1	1	2	1
CO-5	-	-	-	-	-	-	-	1	2	1	3	1	1	1	1

UNIT-I:

Introduction to Economics: Definition, nature, scope and types of Economics. National Income (NI) & types of Inflation.

Forms of Organizing Private and Public-Sector Business Enterprises:

Private Sector Business Enterprises: (i) Sole Proprietorship – Definition, features, merits, limitations & suitability. (ii) Partnership – Definition, Partnership Act, features, types,

merits, limitations, suitability. (iii) Joint-Stock Company – Definition, Companies Act, features, types, merits, limitations, suitability.

Public Sector Business Enterprises: Definition, features, objectives, merits, problems

UNIT-II:

Business Economics: Definition, nature and scope, linkages with other disciplines.

Demand Analysis: Law of Demand, Factors affecting demand; Elasticity of Demand-Types Measurement, Factors affecting and Significance,

Demand Forecasting: Characteristics of Good Demand Forecasting, Steps in Demand Forecasting, Methods of Demand Forecasting.

Supply Analysis: Determinants of Supply, Supply function and Law of Supply.

UNIT-III:

Production, Cost, Market Structures & Pricing:

Production Analysis: Factors of Production, Production Function, Production Function with one variable input, two variable inputs, Returns to Scale, Different Types of Production Functions - Cobb-Douglas.

Cost Analysis: Types of Costs, Short run and long run Cost Functions.

Market Structures: Nature of Competition, Features of Perfect competition, Monopoly, Oligopoly, Monopolistic Competition.

Pricing: Types of Pricing, Product Life Cycle based Pricing, Break Even Analysis (Simple problems)

UNIT-IV:

Introduction to Financial Accounting: Definition, basic principles and double-entry book-keeping, practice of accounting process-Journal, ledger, trial balance and final accounts (simple problems)

UNIT-V:

Ratio Analysis: Meaning, computation of ratios (i) Liquidity Ratios: Current Ratio and Quick Ratio, (ii) Solvency Ratios: Interest Coverage Ratio and Debt-Equity Ratio, (iii) Activity Ratios: Stock/Inventory Turnover Ratio and Debt Turnover Ratio, (iv) Profitability Ratios: Gross Profit Ratio, Net Profit Ratio & Earning Per Share (EPS) Ratio

TEXT BOOKS:

1. Managerial Economics, D. M. Mithani, 9th Edition, Himalaya Publishing House, 2022
2. Managerial Economics, Satya P. Das & J. K. Goyal, 2nd Edition, Sage Publications, 2022
3. Financial Accounting, S. N. Maheswari, 6th Edition, Vikas Publications, 2018

REFERENCES:

1. Managerial Economics, Dominick Salvatore, Siddhartha K. Rastogi, 9th Edition, Oxford Publications, 2020
2. Financial Accounting for Management: An Analytical Perspective, Ambrish Gupta, 6th Edition, Pearson Education, 2018
3. Business Economics, H. L. Ahuja, 13th Edition, S. Chand, 2019
4. Principles of Marketing: A South Asian Perspective, Kotler Philip, Gary Armstrong, Prafulla Y. Agnihotri, and Eshan-ul Haque, 13th Edition, Pearson Education/Prentice Hall of India, 2010

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC1EC401) MICROWAVE ENGINEERING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: EM Waves and Transmission Lines, Antennas and Wave Propagation

COURSE OBJECTIVES:

- To know the required fundamentals on Microwave Transmission lines
- To understand various coupling techniques in waveguides
- To understand the multiport junction concepts and their functionality
- To build a Microwave test bench setup for measurements using Microwave components

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the characteristics of microwave transmission lines

CO-2: Understand the basics of wave propagation inside waveguides

CO-3: Illustrate the functioning of microwave components and Junctions

CO-4: Classify various microwave sources

CO-5: Measure the microwave parameters

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	3	3	-	-	-	-	-	-	-	-	-	3	-	-
CO-2	2	2	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-3	3	3	3	-	-	-	-	-	-	-	-	-	3	-	-
CO-4	2	2	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-5	3	3	2	-	-	-	-	-	-	-	-	-	3	-	-

UNIT-I:

Microwave Transmission Lines: Introduction, Microwave Spectrum and Bands, Applications of Microwaves. Rectangular Waveguides: Solution of Wave Equations in Rectangular coordinates. TE/TM mode Analysis, Expression for E and H fields, Characteristic Equation and Cut-off Frequencies. Phase and Group Velocities, Wavelengths and Impedance Relations.

Microstrip Lines: Introduction, Quasi TEM mode, Z₀ Relations, Effective Dielectric Constant, Applications, Q Factor and Losses.

UNIT-II:

Waveguide Components-I: Cavity Resonators: Introduction, Rectangular Cavities, Dominant Modes and Resonant Frequencies, Q factor and Coupling Coefficients, Illustrative Problems.

Coupling Mechanisms: Probe, Loop, Aperture types. Waveguide Discontinuities: Waveguide Windows, Tuning Screws and Posts, types of Matched Loads.

Waveguide Attenuators: Different types, Resistive Card and Rotary vane Attenuators; Waveguide Phase shifters: Types, Dielectric and Rotary vane Phase shifters.

UNIT-III:

Waveguide Components-II: Ferrite materials –Composition and Characteristics, Faraday Rotation, Ferrite Components - Isolator, Circulator, Gyrator. Scattering matrix: Significance, scattering parameters, formulation and properties of S- Matrix. Waveguide multiport Junctions: E- plane, H-Plane and Magic Tee; Directional coupler –two-hole, Bethe hole types. Calculation of S matrix for multiport junctions.

UNIT-IV:

Microwave Tubes: Limitations of Conventional tubes at Microwave frequencies, Microwave Tubes- Classifications, Two cavity Klystrons–Structure, Velocity Modulation process and Applegate diagram, Reflex Klystrons-Structure, Velocity Modulation, Applegate diagram and Principle of Working, Mode Characteristics ; Travelling Wave tubes, Slow wave structures, Amplification Process, Microwave crossed field tube, Cavity Magnetron-Structure and characteristics, PI mode operation

UNIT-V:

Microwave Solid State Devices: PIN diode and its applications

Transferred Electronic Devices: Introduction, Gunn Diode-Principle, Two valley theory, High field domain, Basic modes of operation.

Avalanche Transit Time Devices: Introduction, Avalanche multiplication. IMPATT, TRAPATT -Principle of Operation.

Microwave Measurements: Description of microwave bench, measurement of Attenuation, low-power, medium and high-power measurements, Bolometer, Frequency, VSWR and Impedance measurements, Antenna radiation pattern measurement, Introduction to reflectometer, Network Analyser.

TEXT BOOKS:

1. Microwave Devices and Circuits, Samuel Y. Liao, 3rd Edition, Pearson, 2003
2. Microwave and Radar Engineering, M. Kulkarni, 5th Edition, Umesh Publications, 2003

REFERENCES:

1. Foundations for Microwave Engineering, R. E. Collin, IEEE Press, 2nd Edition, John Wiley, 2001
2. Microwave Engineering Passive Circuits, Peter A. Rizzi, Prentice Hall of India, 1999
3. Microwave Engineering, David Pozar, 3rd Edition, Addison Wesley 1997
4. Microwave Principles, Herbert J. Reich, J. G. Skalnik, P. F. Ordung and H. L. Krauss, CBS Publishers and Distributors, 2004

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EC401) CELLULAR AND MOBILE COMMUNICATIONS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Analog and Digital Communications, Computer Communication Networks

COURSE OBJECTIVES:

- To learn the basics of mobile telecommunication system
- To be familiar with the Cellular network protocols
- To learn mobile network and transport layers

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply the concept of Cell and Cell Capacity

CO-2: Design a basic 4G cellular network

CO-3: Apply knowledge of LTE-A Cellular networks

CO-4: Create Applications based on Mobile Internet Protocol

CO-5: Establish new protocols using Mobile Transmission Control Protocol

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-2	3	2	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-3	3	3	3	-	-	-	-	-	-	-	-	-	3	-	-
CO-4	3	2	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-5	3	2	2	-	-	-	-	-	-	-	-	-	2	-	-

UNIT-I:

History of Cellular Communications Systems: Cellular Standards - From 1G to 4G and their comparisons, different 4G Standards.

Cellular Concepts: Introduction, Frequency reuse, Channel Assignment and Handoff Strategies, Interference & System capacity, Trunking and Grade of Service, Improving coverage and capacity.

UNIT-II:

Evolution to Fourth-Generation Mobile Cellular Communications: Introduction to 4G , Technologies for 4G - Multi-Input Multi-Output, Multi-User MIMO, Orthogonal Frequency Division Multiplexing, OFDM Multiple Access.

UNIT-III:

Long Term Evolution - Advanced: Frequency Bands, Key Features, Channels, Architecture, Evolved Packet Core, Physical Layer Enhancements - Enhanced

Spectrum Efficiency, Improved Interference Management, High Data Rates, Coverage Enhancements and LTE-A Radio Interface Protocols.

UNIT-IV:

Mobile Network Layer:

Mobile IP - Goals, assumptions and requirements, Entities and terminology, IP packet delivery, Agent discovery, Registration, Tunnelling and encapsulation, Optimizations, Reverse tunnelling, IP micro-mobility support,
Routing - Destination sequence distance vector, Dynamic source routing, Alternative metrics, Overview of ad-hoc routing protocols.

UNIT-V:

Mobile Transport and Application Layer: Classical TCP improvements - Indirect TCP, Snooping TCP, Mobile TCP, Fast retransmit/fast recovery, Transmission/time-out freezing, Selective retransmission, Transaction-oriented TCP
Wireless Application Protocol - Architecture, Wireless Datagram Protocol, Wireless Transport Layer Security, Wireless Transaction Protocol, Wireless Session Protocol, Wireless application environment, Wireless markup language, Wireless Markup Language Script.

TEXT BOOKS:

1. Cellular Communication Networks and Standards The Evolution from 1G to 6G, Wei Jiang, Bin Han, Springer
2. Wireless Communications Principles & Practice, Theodore S. Rappaport, 2nd Edition, Pearson Education.
3. Mobile Communications, Jochen Schiller, 2nd Edition, Prentice Hall of India, 2003

REFERENCES:

1. Fundamentals of Mobile Computing, Prasant Kumar Pattnaik, Rajib Mall, Prentice Hall of India, 2012
2. Introduction to Wireless and Mobile Systems, Dharma Prakash Agarwal, Qing and An Zeng, Thomson Asia, 2005
3. Principles of Mobile Computing - II, Uwe Hansmann, Lothar Merk, Martin S. Nicklons and Thomas Stober, Springer, 2003
4. Mobile Cellular Telecommunications-Analog and Digital Systems, William C. Y. Lee, 2nd Edition, Tata McGraw-Hill, 2006
5. AdHoc Mobile Wireless Networks, C. K. Toh, 1st Edition, Pearson Education, 2002

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EC402) DSP PROCESSORS AND ARCHITECTURES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Digital Signal Processing, Microprocessors and Microcontrollers

COURSE OBJECTIVES:

- To study the architectural features of DSP Processor
- To learn about numeric formats and sources of errors in DSP implementation
- To know the basic building blocks of DSP Processor
- To understand the code optimization techniques

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: understand the architecture and evaluate the finite word length effects

CO-2: Apply the pipelining concepts and evaluate the computational accuracy

CO-3: Understand the basic building blocks of DSP processor

CO-4: Implement the real time application using DSP processor

CO-5: Apply the code optimization techniques

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-2	3	2	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-3	3	2	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-4	3	2	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-5	3	2	2	-	-	-	-	-	-	-	-	-	2	-	-

UNIT-I:

Introduction to DSP Processors: Digital Signal Processors, VLIW and SHARC Architecture, Comparison of DSP processors with microprocessors.

Implementation considerations : Data representations and arithmetic, Introduction to finite word length effects, Input quantization, Coefficient quantization, overflow and solutions, rounding and truncation, real time implementation considerations.

UNIT-II:

Execution Control and Pipelining: Hardware looping, Interrupts, Stacks, Relative Branch support, Pipelining and Performance, Pipeline Depth, Interlocking, Branch effects, Interrupt effects, Pipeline Programming models.

Computational Accuracy in DSP Implementations: Number formats for signals and coefficients in DSP system, Dynamic range and precision, sources of errors in DSP implementation, A/D conversion errors, DSP computational errors, D/A conversion errors.

UNIT-III:

Architecture for programmable DSP Devices: Basic Architectural features, DSP computational building blocks, bus architecture and memory, Data and addressing capabilities, Address generation unit, programmability and program execution, speed issues, Features for external interfacing.

UNIT-IV:

Architecture and Instruction Set of the C6x Processor: TMS320C6x Architecture, Functional Units, Fetch and Execute Packets, Pipelining, Registers, Linear and Circular Addressing Modes, TMS320C6x Instruction Set

On-chip peripherals: Timers, Interrupts, Interrupt Control Registers, Interrupt Acknowledgment, Multichannel Buffered Serial Ports, Direct Memory Access, Memory Considerations

UNIT-V:

Code Optimization: Optimization Steps, Compiler Options, Intrinsic C Functions, Procedure for Code Optimization, Programming Examples Using Code Optimization Techniques

TEXT BOOKS:

1. Digital Signal Processing, Avtar Singh and S. Srinivasan, Thomson Publications, 2016
2. Digital Signal Processing - A Practical Approach, Emmanuel C. Ifeachor, Barrie W. Jervis, 2nd Edition, Pearson Publications. 2002
3. Digital Signal Processing and Applications with the C6713 and C6416 DSK, Rulph Chassaing, Wiley, 2005

REFERENCES:

1. Digital Signal Processors Architectures, Implementations and Applications, Sen M. Kuo, Woon-Seng S. Gan, Pearson Publications, 2009
2. Digital Signal Processors, Architecture, Programming and Applications, B. Venkata Ramani and M. Bhaskar, Tata McGraw-Hill, 2007
3. DSP Processor Fundamentals, Architectures and Features, Lapsley, S. Chand, 2003
4. DSP Applications with TMS 320 Family, K. Shin, Prentice Hall, 1987

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EC403) LANGUAGES FOR EMBEDDED SOFTWARE

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Microprocessor and Microcontrollers, Programming for Problem Solving

COURSE OBJECTIVES:

- To understand the key characteristics of programming languages for embedded software development
- To develop advanced C programming skills for embedded systems
- To understand the enhancements provided by C++
- To develop proficiency in Python programming for embedded systems

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Identify and select appropriate programming language and techniques for embedded system design

CO-2: Familiarize with embedded C in programming practical embedded applications

CO-3: Understand the object-oriented programming principles in C++

CO-4: Apply object-oriented programming principles in C++ to design embedded software

CO-5: Utilize Python programming to develop embedded system development

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	3	2	3	2	2	1	1	-	-	3	3	3	3
CO-2	3	2	3	2	3	2	2	1	1	-	-	3	3	3	3
CO-3	3	2	3	2	2	2	2	1	1	-	-	3	3	3	3
CO-4	3	2	2	2	3	2	2	1	1	-	-	3	3	3	3
CO-5	2	2	3	2	3	2	2	1	1	-	-	3	3	3	3

UNIT-I:

Embedded Programming: Programming Languages - Desired Language Characteristics, Data typing, Control, Multitasking and Task scheduling, Timing specifications, Run-time exception handling Programming in Assembly language, Use of High-Level languages – Structural programming, Object-oriented programming.

UNIT-II:

Embedded C Programming: Bitwise operations, Dynamic memory allocation, Linked stack and queue, Sparse matrices, Binary tree, Writing drivers for LEDs, Embedded Software Development Life Cycle: Overview of Waterfall and Agile methodologies.

UNIT-III:

C++ Programming-I: C++ enhancements to C, objects, classes, data members & methods, 'cin', 'cout', constructors, destructors, 'this' pointer, formatting and I/O manipulators, dynamic memory allocation, abstraction, encapsulation.

UNIT-IV:

C++ Programming-II: Need of operator overloading, overloading using friends, type conversions, base and derived classes, friend classes, inheritance, types of inheritance, virtual base class, polymorphism, virtual functions.

Function and class templates for generic programming, Exception handling: Try-catch-throw blocks for error management.

UNIT-V:

Embedded Python: PYTHON-syntax, statements, functions, Built-in-functions and Methods, Modules in PYTHON, Exception Handling: Need for Exception handling, Raising exceptions.

TEXT BOOKS:

1. C.M. Krishna, Kang G. Shin, Real-Time Systems, McGraw-Hill International, 1997
2. Core Python Programming, Chun, Pearson Education, Prentice Hall Professional, 2001
3. C How to Program: With an Introduction to C++, Paul Deitel, Harvey Deitel, Pearson Education, 2016

REFERENCES:

1. Embedded Systems-Architecture, Programming and Design, Raj Kamal, Tata McGraw-Hill, 2011
2. Embedded C, Michael J. Pont, 2nd Edition, Pearson Education, 2008
3. C++ for Embedded Systems, Arkadiusz Mazur
4. Fundamentals of Embedded Software where C and Assembly Meet, Daniel W. Lewis, Pearson Education

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EC404) VLSI SCRIPTING LANGUAGES

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Programming for Problem Solving

COURSE OBJECTIVES:

- To understand the importance of scripting languages in VLSI Design
- To describe the various PERL concepts used in VLSI design for large data handling
- To understand utilization of TCL in CAD tools interfacing

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Gain fluency in programming with scripting languages

CO-2: Create and run scripts using PERL

CO-3: Demonstrate the use of PERL in developing systems and web applications

CO-4: Develop the applications using TCL scripting

CO-5: Learn the advanced programming skills of TCL

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	1	2	2	2	-	-	-	1	1	1	1	-	3	2
CO-2	1	1	2	2	2	-	-	-	1	1	1	1	-	3	2
CO-3	1	1	2	2	2	-	-	-	1	1	1	1	-	3	2
CO-4	1	1	2	2	2	-	-	-	1	1	1	1	-	3	2
CO-5	1	1	2	2	2	-	-	-	1	1	1	1	-	3	2

UNIT -I:

Basics of Linux: Linux Architecture, shells, Files and Directories: file systems, system calls for the file I/O Operations, file permissions & ownership, creating, removing, and changing directories, GREP, SED, AWK.

Introduction to Scripting Languages: Origin of Scripting languages, Characteristics of scripting languages, uses of scripting languages.

UNIT-II:

PERL: Introduction to PERL, Names and values, Variables and assignment, Scalar expressions, Control structures, Built-in functions, Collections of Data, Working with arrays, Lists and hashes, Simple input and output, Strings, Patterns and regular expressions, Subroutines, Scripts with arguments.

UNIT-III:

Advanced PERL: Finer points of Looping, Subroutines, Using Pack and Unpack, Working with files, Type globs, Eval, References, Data structures, Packages, Libraries and modules, Objects, Objects and modules in action, Tied variables

UNIT-IV:

TCL: The TCL phenomena, Philosophy, Structure, Syntax, Parser, Variables and data in TCL, Control flow, Data structures, Simple input/output, Procedures, Working with Strings, Patterns, Files and Pipes, Example code.

UNIT-V:

Advanced TCL: The eval, source, exec and up-level commands, Libraries and packages, Namespaces, Trapping errors, Event-driven programs, Making applications 'Internet-aware', 'Nuts-and-bolts' internet programming, Security issues

TEXT BOOKS:

1. The World of Scripting Languages, David Barron, Wiley Student Edition, 2010
2. TCL/TK: A Developer's Guide, Clif Flynt, Morgan Kaufmann Series, 2003
3. Learning Perl, Randal L. Schwartz, 6th Edition, O' Reilly, 2011

REFERENCES:

1. Linux: The Complete Reference, Richard Peterson, 6th Edition, McGraw-Hill, 2008
2. Programming Perl, Tom Christiansen, brian d foy, Larry Wall, Jon Orwant, Shroff Publishers, 4th Edition, 2012
3. Think Perl 6: How to Think Like a Computer, Allen Downey and Laurent Rosenfeld, O'Reilly, 2017
4. Field Programmable Gate Arrays, John V. Old Field, Richrad C. Dorf, Wiley, 2008

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PE1EC405) ORGANIC ELECTRONICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Electronic Devices and Circuits

COURSE OBJECTIVES:

- To understand the scope of organic and molecular electronics
- To understand the electrical conductivity at low and high frequencies
- To study the phenomena of organic devices
- To learn electroactive organic compounds
- To analyze the architecture, characterization, and utilization of electronic components

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the basics of organic electronics

CO-2: Analyze electrical conductivity at low and high frequencies

CO-3: Understand the structure and properties of organic devices

CO-4: Explain energy bands, charge transport, and doping in organic electronic materials

CO-5: Exemplify the architecture, characterization, and utilization of electronic components based on organic electronic materials

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	2	1	3	3	-	-	-	2	2	3	-	-
CO-2	3	3	3	2	2	2	1	-	-	-	-	-	3	-	-
CO-3	3	2	3	2	1	1	1	-	-	-	-	2	3	-	-
CO-4	3	2	3	2	1	1	1	-	-	-	-	2	3	-	-
CO-5	3	2	3	2	1	1	1	-	-	-	-	2	3	-	-

UNIT-I:

Material's Foundations: Introduction to organic and molecular electronics, Organic materials for electronics, molecular electronics, Evolution, Morre's law, Electronic Structure, Chemical Bonding, Bonding in Organic Compounds, Crystalline and nanocrystalline materials, Polymers, Diffusion.

UNIT-II:

Electrical Conductivity: Classical Theory, Energy Bands in Solids, Organic Compounds, Low-frequency conductivity, Conductivity at high frequencies.

UNIT-III:

Optical Phenomena: Electromagnetic radiation, refractive index, Interaction of EM waves with Organic Molecules, Transmission and Reflection from Interfaces, Waveguiding, Surface Plasmons, Photonic Crystals.

UNIT-IV:

Electroactive Organic Compounds: Selected Topics in Chemistry, Conductive Polymers, Charge Transfer Complexes, Graphene, Fullerenes and Nanotubes, Piezoelectricity, Magnetic Materials.

Tools for Molecular Electronics, Liquid Crystals and Devices.

UNIT-V:

Plastic Electronics: Organic Diodes, MIS structures, Organic FETs, Transparent Conducting Films, Organic LEDs, Organic Photovoltaic Devices. Chemical Sensors and Physical Actuators; Molecular and Nanoscale Electronics; Bioelectronics.

TEXT BOOK:

1. Organic and Molecular Electronics: From Principles to Practice, Michael C. Petty, 2nd Edition, Wiley, 2018

REFERENCES:

1. Organic Electronics: Foundations to Applications, Stephen R. Forrest, 1st Edition, Oxford University Press, 2020
2. Organic Flexible Electronics: Fundamentals, Devices and Applications, Piero Cosseddu and Mario Caironi, 1st Edition, Elsevier, 2021

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2EC401) MICROWAVE ENGINEERING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: EM Waves and Transmission Lines

COURSE OBJECTIVES:

- To study the performance of microwave sources
- To calculate scattering parameters of microwave junctions
- To measure the characteristic parameters of Microwave components

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Analyze the performance characteristics of Microwave sources

CO-2: Analyze the scattering parameters of microwave junctions

CO-3: Understand the design aspects of Microwave components

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	-	2	-	-	-	2	2	2	-	2	-	-
CO-2	3	3	2	-	2	-	-	-	2	2	2	-	2	-	-
CO-3	2	3	3	-	3	-	-	-	2	2	2	-	2	-	-

LIST OF EXPERIMENTS:

1. Reflex Klystron Characteristics.
2. Gunn Diode Characteristics.
3. Attenuation Measurement.
4. Directional Coupler Characteristics.
5. VSWR Measurement.
6. Impedance, Frequency and Waveguide parameters Measurement.
7. Scattering parameters of Circulator.
8. Scattering parameters of E-Plane Tee and H-Plane Tee junctions.
9. Scattering parameters of Magic Tee.
10. Design and analysis of a Microstrip patch antenna using HFSS tool.
11. Design and analysis of rectangular waveguide using HFSS tool.
12. Design and analysis of microstrip line using HFSS tool.

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VII Semester

(22PC2EC304) MACHINE LEARNING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Artificial Intelligence and Machine Learning, Python Programming and Practice

COURSE OBJECTIVES:

- To learn foundational machine learning techniques
- To implement advanced algorithms like CNNs and RNNs
- To develop hands-on experience deploying ML models on edge devices
- To explore practical applications of machine learning

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Develop and implement foundational machine learning models

CO-2: Apply deep learning techniques including CNNs and RNNs

CO-3: Develop applications using machine learning and deep learning

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	1	-	-	-	-	2	2	-	-	1	3	-
CO-2	2	2	2	1	-	-	-	-	2	2	-	-	1	3	-
CO-3	2	2	2	1	-	-	-	-	2	2	-	-	1	3	-

LIST OF PROGRAMS:

1. Implement Linear Regression.
2. Implement Logistic Regression.
3. Implement KNN classification and regression.
4. Implement Support Vector Machine Classifier.
5. Implement Decision Tree.
6. Implement Random Forest.
7. Implement Naive Bayes algorithm for the given data set.
8. Implement K-means and Hierarchical Clustering.
9. Implement CNN for image classification.
10. Implement RNN and LSTM for time series data.
11. Implement an algorithm for object detection.
12. Implement an algorithm for Facial recognition.

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC406) WIRELESS COMMUNICATIONS AND NETWORKS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Analog and Digital Communications, Computer and Communication Networks

COURSE OBJECTIVES:

- To build understanding of the fundamental concepts of wireless communications
- To learn mathematical modelling of a wireless communication channel
- To understand the fundamentals of a basic wireless LAN network
- To differentiate fixed IP and Mobile IP

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand basic principles of design of Wireless system

CO-2: Differentiate among various multiple access techniques for Wireless communications

CO-3: Analyse issues in various wireless propagation models

CO-4: Understand the design of a wireless communication system

CO-5: Analyse design issues of WLAN network standard

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-2	2	3	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-3	2	3	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-4	3	3	3	-	-	-	-	-	-	-	-	-	2	-	-
CO-5	2	3	3	-	-	-	-	-	-	-	-	-	2	-	-

UNIT-I:

Introduction to Wireless Communication Systems: Evolution, Examples and comparison of common wireless communication systems, trends in cellular radio and personal communications.

Modern Wireless Communication Systems: Third generation wireless networks, wireless local loop and Local Multipoint Distribution Service, wireless local area networks, Bluetooth and personal area networks.

UNIT-II:

Multiple Access Techniques for Wireless Communication: Frequency Division Multiple Access, Time Division Multiple Access, Spread Spectrum Multiple Access - Frequency Hopping and Code Division, Space Division Multiple Access.

UNIT-III:

Mobile Radio Propagation - Large-Scale Path Loss: Introduction to Radio Wave Propagation, Free Space Propagation Model, Relating Power to Electric Field, Propagation Mechanisms - Reflection, Ground Reflection (Two-Ray) Model, Diffraction, Scattering.

Outdoor Propagation Models - Longley-Ryce, Okumura and Hata Model.

Indoor Propagation Models - Partition losses, Partition losses between Floors, Log-distance path loss.

UNIT-IV:

Mobile Radio Propagation - Small-Scale Fading and Multipath: Small Scale Multipath propagation, Impulse Response Model of a multipath channel, Parameters of Mobile Multipath Channels-Time Dispersion Parameters, Coherence Bandwidth, Doppler Spread and Coherence Time.

Types of Small-Scale Fading: Fading effects Due to Multipath Time Delay Spread, Flat fading, Frequency selective fading, Fading effects Due to Doppler Spread-Fast fading, slow fading.

UNIT-V:

Wireless LAN Technology: Infrared LANs, Spread spectrum LANs, Narrow band microwave LANs, IEEE 802 protocol Architecture, IEEE 802.11 architecture, services, Medium access control and physical layer.

Blue Tooth: Overview, Radio, Base band and Links manager Specification, Logical link control and adaptation protocol.

TEXT BOOKS:

1. Wireless Communications -Principles Practice, Theodore S. Rappaport, 2nd Edition, Prentice Hall of India, 2002
2. Wireless Communications and Networks, William Stallings, 2nd Edition, Pearson Education, 2009
3. Principles of Wireless Networks -A Unified Approach, Kaveh PahLaven, Prashanth Krishna Murthy, Pearson Education, 2007

REFERENCES:

1. Wireless Digital Communications, Dr. Kamilo Feher, Prentice Hall of India, 2003
2. Mobile Communications, Jochen Schiller, 2nd Edition, Pearson Education, 2009
3. Wireless Communications, Andreas F. Molisch, Wiley India, 2006

B.Tech. VIII Semester

(22PE1EC407) SPEECH SIGNAL PROCESSING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Signals and Systems, Digital Signal Processing

COURSE OBJECTIVES

- To understand speech production and related parameters of speech
- To learn the time domain analysis of speech signal
- To understand the spectral analysis of speech and modeling
- To learn the enhancement and coding of speech signal
- To study speech recognition and synthesis system

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the basic organs and parameters of speech

CO-2: Analyze the speech signal in time domain

CO-3: Probe the speech spectrum and modelling

CO-4: Enhance and code the speech signal

CO-5: Design a speech recognition and synthesis system

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	1	-	-	-	-	-	-	-	-	-	3	-	-
CO-2	3	3	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-3	3	3	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-4	3	3	3	-	-	-	-	-	-	-	-	-	3	-	-
CO-5	3	2	3	-	-	-	-	-	-	-	-	-	3	-	-

UNIT-I:

Fundamentals of Speech Production and Processing: Anatomy and physiology of speech organs, the process and acoustic theory of speech production, uniform lossless tube model, effect of losses in vocal tract, effect of radiation at lips, digital models for speech signals.

UNIT-II:

Time Domain Methods for Speech Processing: Time domain parameters of speech signal, methods for extracting the speech parameters, short time auto correlation function, pitch period estimation using auto correlation function.

UNIT-III:

Frequency Domain Methods for Speech Processing: Short-time Fourier transform (STFT), sampling the STFT in time and frequency, the speech spectrogram, homomorphic

systems for convolution, definition of the Cepstrum and complex Cepstrum, pitch extraction using homomorphic speech processing.

Speech Modeling: Introduction to Markov Processes, Hidden Markov Model development and Evaluation, Optimal State Sequence Viterbi Search algorithm, Baum-Welch Parameter Re-estimation.

UNIT-IV:

Speech Enhancement: Introduction, Single Microphone Approach, Spectral Subtraction, Enhancement by re-synthesis, Comb filter, Wiener filter, Multi Microphone Approach.

Speech Coding: Predictive Coding, Delta Modulation, Adaptive Differential PCM Systems, Multi-Pulse Excitation Linear Prediction (MPLP), Code-Excited Linear Prediction (CELP).

UNIT-V:

Speech Recognition: Architecture of large vocabulary continuous speech recognition system, acoustics and language models

Speech Synthesis: Concatenative and waveform synthesis methods, Intelligibility and naturalness.

TEXT BOOKS:

1. Introduction to Digital Speech Processing, Lawrence R. Rabiner and Ronald W. Schafer, now Publishers Inc., 2007
2. Speech Enhancement, Philipos C. Loizou, 2nd Edition, CRC Press, Taylor & Francis, 2013
3. Fundamentals of Speech Recognition, Lawrence Rabiner and Biing-Hwang Juang, Pearson Education, 2003

REFERENCES:

1. Digital Processing of Speech Signals, L. R. Rabiner and R.W. Schafer, Prentice-Hall, 1978
2. Speech Communications Human and Machine, Douglas O. Shaughnessy, 2nd Edition, IEEE Press, 2000
3. Statistical Methods of Speech Recognition, Frederick Jelinek, MIT Press, 1997
4. Speech Recognition, Claudio Becchetti and Lucio Prina Ricotti, John Wiley and Sons, 1999
5. Discrete Time Speech Signal Processing: Principles and Practice, Thomas F. Quateri, Pearson Education, 2004

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC408) RADAR SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: EM Waves and Transmission Lines, Microwave Engineering

COURSE OBJECTIVES:

- To understand the fundamental concepts of a radar system
- To know the various types of radar systems
- To understand the concepts of MTI and tracking radars
- To understand radar measurements

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the principle of radar and factors affecting its performance

CO-2: Analyse different types of radar systems and assess their performance

CO-3: Know the characteristics of MTI radars and their performance

CO-4: understand the principles of tracking radars

CO-5: Analyze the radar receiver performance parameters

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-2	3	2	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-3	3	2	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-4	3	2	2	-	-	-	-	-	-	-	-	-	2	-	-
CO-5	2	2	2	-	-	-	-	-	-	-	-	-	2	-	-

UNIT-I:

Basics of Radar: Introduction, Radar block diagram and operation, Maximum Unambiguous Range, Simple form of Radar Equation, Radar frequencies and Applications. Prediction of Range Performance, Minimum detectable Signal, Receiver Noise, Modified Radar Range Equation, Illustrative Problems.

Radar Equation: SNR, Envelope Detector, False Alarm Time and Probability, Integration of Radar Pulses, Radar Cross Section of Targets (simple targets: sphere, cone-sphere), Transmitter Power, PRF and Range Ambiguities. Systems Losses (qualitative treatment) Illustrative Problems.

UNIT-II:

CW and Frequency Modulated Radar: Doppler Effect, CW Radar – Block Diagram, Isolation between Transmitter and receiver, Non-zero IF Receiver, Receiver Bandwidth Requirements, Applications of CW Radar. Illustrative problems.

FM-CW Radar: Range and Doppler Measurement, Block Diagram and Characteristics (Approaching/ Receding Targets), FM-CW altimeter, Multiple Frequency CW Radar.

UNIT-III:

MTI and Pulse Doppler Radar: Introduction, Principle, MTI Radar with - Power Amplifier Transmitter and Power Oscillator Transmitter, Delay Line Cancellers – Filter Characteristics, Blind Speeds, Double Cancellation, Staggered PRFs. Range Gated Doppler Filter. MTI Radar Parameters, Limitations to MTI Performance, MTI versus Pulse Doppler Radar.

UNIT-IV:

Tracking Radar: Tracking with Radar, Sequential Lobing, Conical Scan, Monopulse Tracking Radar – Amplitude Comparison Monopulse (one- and two- coordinates), Phase Comparison Monopulse. Tracking in Range, Acquisition and Scanning Patterns. Comparison of Trackers.

UNIT-V:

Detection of Radar Signals in Noise: Introduction, Matched Filter Receiver – Response Characteristics and Derivation, Correlation Function and Cross-correlation Receiver, Efficiency of Non-matched Filters, Matched Filter with Non-white Noise. Radar Receivers: Noise Figure and Noise Temperature. Displays – types. Duplexers – Branch type and Balanced type.

TEXT BOOKS:

1. Introduction to Radar Systems, Merrill I. Skolnik, 2nd Edition, Tata Mcgraw-Hill, 2007
2. Radar Principles, Peebles Jr. P. Z., Wiley, 1998

REFERENCES:

1. Introduction to Radar Systems, Merrill I. Skolnik, 3rd Edition, Tata Mcgraw-Hill, 2001
2. Radar: Principles, Technology, Applications, Byron Edde, Pearson Education, 2004

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC409) INTRODUCTION TO ASIC DESIGN

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Switching Theory and Logic Design, CMOS VLSI Design

COURSE OBJECTIVES:

- To prepare the student to be an entry-level industrial standard ASIC designer
- To learn simulation and synthesis steps in ASIC design
- To understand physical design flow for ASIC
- To know various stages of physical design
- To make the students proficient in verifying and testing ASIC design

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the various types of ASICs and logic cells involved in ASIC design

CO-2: Analyze simulation, synthesis and timing issues in ASIC design flow

CO-3: Understand and Analyze Partitioning and floor planning steps in ASIC Design

CO-4: Analyze placement and routing steps in ASIC Design

CO-5: Identify various faults exist in VLSI circuits and importance of testing

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	3	-	-	-	-	-	-	-	-	-	-	3	3
CO-2	2	2	3	-	-	-	-	-	-	-	-	-	-	2	2
CO-3	2	2	3	-	2	-	-	-	-	-	-	-	-	3	3
CO-4	2	2	3	-	-	-	-	-	-	-	-	-	-	2	2
CO-5	2	2	3	-	1	-	-	-	-	-	-	-	-	2	2

UNIT-I:

Introduction to ASICs:

Types of ASICs: Full custom ASICs and Standard cell based ASICs, Gate array based ASICs, programmable Logic Devices, FPGAs, ASIC Design flow, Combinational Logic Cell: Pushing Bubbles, Drive strength, Transmission Gates, Exclusive-OR Cells, Sequential logic cell: Latch, Flip-Flop, Clocked Inverter, Data path logic cells: Ripple carry adder, I/O cell.

UNIT-II:

Simulation and Synthesis: Simulation, types of simulation- Gate-level Simulation, Switch-Level simulation, Transistor-Level Simulation, Logic Synthesis: Comparator/Mux example, Static timing Analysis: Timing paths, Clock issues, setup and hold time Violations.

UNIT-III:

Physical Design Flow: Partitioning- System partitioning: Goals and objectives and a simple example of partitioning. Floor-planning – Goals and objectives, Floor planning tools, channel definition, I/O and power planning, Clock planning

UNIT-IV:

Placement: Goals and Objectives of placement, Min cut constructive placement algorithm and iterative improvement placement algorithm.

Routing: Goals and Objectives of global routing, detailed routing - Measurement of Interconnect delay.

UNIT-V:

Testing: Importance of testing faults-Reliability, fault models, physical faults, stuck at fault models, logical faults.

TEXT BOOKS:

1. Application Specific Integrated Circuits, J. Smith, Addison-Wesley, 2010
2. Physical Design Essentials, Khosrow Golshan, Springer, 2007

REFERENCES:

1. Algorithms for VLSI Design Automation, S. H. Gerez, Wiley, 2006
2. Algorithms for VLSI Physical Design Automation, Naveed Sherwani, 3rd Edition, 2013

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC410) APPLICATION OF DATA ANALYTICS IN IOT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: IoT Architectures and Applications

COURSE OBJECTIVES:

- To understand the concepts of the Internet of Things (IoT) and data analytics
- To learn about cloud architecture and types of clouds
- To study about cloud platforms
- To understand the principles of IoT health data
- To study about the healthcare applications of IoT data analytics

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the core components and architecture of IoT and data analytics

CO-2: Design the cloud architecture

CO-3: Analyze different cloud platforms

CO-4: Apply the analytics for IoT healthcare

CO-5: Develop a Healthcare application

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	-	-	-	-	-	-	-	-	-	-	-	3
CO-2	2	2	2	-	-	-	-	-	-	-	-	-	-	-	3
CO-3	2	2	2	-	-	-	-	-	-	-	-	-	-	-	3
CO-4	2	2	2	-	-	-	-	-	-	-	-	-	-	-	3
CO-5	2	2	2	-	-	-	-	-	-	-	-	-	-	-	3

UNIT-I:

IoT Analytics and Challenges: Introduction to IoT Analytics, Challenges, Business Value Concerns, IoT Devices, IoT Analytics for the Cloud, Cloud Security and Analytics.

Virtualization: Virtual machines, Implementation Levels of Virtualization -Virtualization Structures/Tools and Mechanisms-Virtualization of CPU, Memory, and I/O Devices

UNIT-II:

Cloud Computing and Service Models: Public, Private and Hybrid Clouds, Infrastructure as a Service Providers, Platform as a Service and Software as a Service . Architectural Design of Compute and Storage Clouds: Design, Development and Challenges of Cloud Architecture

UNIT-III:

Cloud Platforms:

AWS: Introduction, Key Concepts, Key core services, key services for IoT analytics.

Microsoft Azure: Introduction, Data Lake Store, Analysis Services, HDInsight
Thing Worx: Introduction, Core, Connection Services, Edge

UNIT-IV:

Health IoT Data Analytics: IoT based Healthcare Architecture, System Design for IoT Healthcare Applications, Sensors and Challenges.

IoT Health Data: Introduction, Characteristics and Types of Health data.

UNIT-V:

Health Data Analytics using IoT: Introduction, Methods and bottlenecks of the health IoT data analytics.

IoT for Healthcare Applications: Introduction, IoT for healthcare, Current Issues and Challenges.

TEXT BOOKS:

1. Analytics for the internet of things, Minter, Andrew, Packt Publishing, 2017
2. Distributed and Cloud Computing: From Parallel Processing to the Internet of Things, Hwang, Kai, Jack Dongarra, and Geoffrey C. Fox, Morgan Kaufmann, 2013
3. IoT-based Data Analytics for the Healthcare Industry: Techniques and Applications, Singh Sanjay Kumar, Ravi Shankar Singh, Anil Kumar Pandey, Sandeep S. Udmale, and Ankit Chaudhary, Academic Press, 2020

REFERENCES:

1. Cloud Computing: Concepts, Technology & Architecture, Erl Thomas, Ricardo Puttini, Zaigham Mahmood, Pearson Education, 2013
2. Internet of Things And Big Data Analytics Toward Next-Generation Intelligence, Dey Nilanjan, Aboul Ella Hassanien, Chintan Bhatt, Amira Ashour, and Suresh Chandra Satapathy, Vol. 35, Berlin, Springer, 2018

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC411) SATELLITE COMMUNICATIONS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Antennas and Wave Propagation, Microwave Engineering

COURSE OBJECTIVES:

- To understand the construction and principles of satellites, earth stations
- To know the tracking techniques of satellites
- To learn about various multiple accessing techniques
- To study about the power budget for satellite links
- To know about the GPS and low earth orbit and geo-stationary satellite systems

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the orbital mechanics

CO-2: Comprehend the functioning of subsystems of a satellite

CO-3: Apply various multiple access techniques

CO-4: Design the power budget for satellite links

CO-5: Reveal the principles of GPS, low earth orbit and geo-stationary satellite systems

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	3	1	-	-	-	-	-	-	-	-	3	-	-
CO-2	3	3	3	1	-	-	-	-	-	-	-	-	3	-	-
CO-3	3	3	3	1	-	-	-	-	-	-	-	-	3	-	-
CO-4	3	3	3	3	-	-	-	-	-	-	-	-	3	-	-
CO-5	3	2	3	1	-	-	-	-	-	-	-	-	2	-	-

UNIT-I:

Introduction: Origin of Satellite Communications, Historical Background, Basic Concepts of Satellite Communications, Frequency allocations for Satellite Services, Applications, Future Trends of Satellite Communications.

Orbital Mechanics and Launchers: Orbital Mechanics, Look Angle determination, Orbital perturbations, Orbital determination, Launches and Launch vehicles, Orbital effects in communication systems performance.

UNIT-II:

Satellite Subsystems: Attitude and Orbit control system, Telemetry, Tracking, Commanding and Monitoring, Power Systems, Communication Subsystems, Satellite antennas, Equipment reliability and Space qualification.

UNIT-III:

Multiple Access: Frequency Division Multiple Access, Intermodulation, calculation of C/N. Time Division Multiple Access, Frame structure, Examples. Satellite Switched TDMA Onboard processing, DAMA, Code Division Multiple Access, Spread Spectrum Transmission and Reception.

UNIT-IV:

Satellite Link Design: Basic transmission theory, system noise temperature and G/T ratio, Design of down links, Uplink design, Design of satellite links for specified C/N, System design examples.

Earth Station Technology: Introduction, Transmitters, Receivers, Antennas, Tracking systems, Terrestrial Interface, Primary Power test methods.

UNIT-V:

Low Earth Orbit and Geo-Stationary Satellite Systems: Orbit considerations, Coverage and Frequency Consideration, Delay and Throughput considerations, Systems considerations, Operational NGSO Constellation Designs.

Satellite Navigation and Global Positioning System: Radio and Satellite Navigation, GPS Position Location principles, GPS Receivers and Codes, Satellite Signal Acquisition, GPS Navigation Message, GPS Signal Levels, GPS Receiver Operation, GPS C/A code accuracy, Differential GPS.

TEXT BOOKS:

1. Satellite Communications, Timothy Pratt, Charles Bostian and Jeremy Allnutt, 2nd Edition, Wiley Publications, 2003
2. Satellite Communications Engineering, Wilbur L. Pritchard, Robert A. Nelson and Henri G. Suyderhoud, 2nd Edition, Pearson Publications, 2003

REFERENCES:

1. Satellite Communications: Design Principles, M. Richharia, 2nd Edition, B. S. Publications, 2003
2. Satellite Communication, D. C. Agarwal, 5th Edition Khanna Publications
3. Fundamentals of Satellite Communications, K. N. Raja Rao, Prentice Hall of India, 2004
4. Satellite Communications, Dennis Roddy, 4th Edition, McGraw-Hill, 2009

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC412) 5G COMMUNICATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Analog and Digital Communications, Computer and Communication Networks

COURSE OBJECTIVES:

- To learn the basic technologies of 5G
- To study the concepts of MIMO, mm Wave
- To understand 5G services
- To learn 5G use cases

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand Fundamentals of 5G Technology

CO-2: Explore Multiple Input Multiple Output Communications

CO-3: Analyse the propagation models for mm Wave

CO-4: Comprehend device to device (D2D) communication and standardization

CO-5: Identify the 5G use cases for various applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	2	2	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-2	2	2	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-3	2	2	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-4	3	2	2	-	-	-	-	-	-	-	-	-	3	-	-
CO-5	3	2	2	-	-	-	-	-	-	-	-	-	3	-	-

UNIT-I:

Fundamentals of 5G Technology: 10 Pillars of 5G, 5G RAN Architecture and Network Slicing.

UNIT-II:

Multiple Input Multiple Output Communications: Spatial Multiplexing, Spatial Diversity, Beamforming in Multiple Input Multiple Output systems, Hybrid Precoding, 5G Communication Landscape, Related work on 5G.

UNIT-III:

Radio Wave Propagation for Mm Wave: Introduction, Large-scale Propagation Channel Effects, Small-Scale Channel Effects, Spatial Characterization of Multipath and Beam Combining, Outdoor Channel Models, Indoor Channel Models.

UNIT-IV:

5G Services: OFDMA-based Cellular Network and underlying D2Ds – D2D Deployment Modes, System Level Performance Evaluation, Signal-to-interference Ratio with Activity, Coverage Probability Analysis and Throughput Analysis.

UNIT-V:

5G Use Cases : Smart Vehicles, Mission Critical Service Using URLLC, Massive Internet-of-Things, Enhanced Mobile Broadband, Interworking with Internet-of-Things Application Servers

TEXT BOOKS:

1. Fundamentals of 5G Mobile Networks Jonathan Rodriguez, Wiley, 2015
2. Fundamentals of 5G Communications: Connectivity for Enhanced Mobile Broadband and Beyond Wanshi Chen, Peter Gaal, Juan Montojo, Haris Zisimopoulos, 2021
3. TMillimetre Wave Wireless Communication, S. Rappaport, R. W. Heath Jr., R. C. Daniels, and J. M. Murdock, Pearson Education, 2015
4. End-to-End Mobile Communications: Evolution to 5G, 1st Edition, Syed S. Husain, 2020

REFERENCES:

1. 5G Mobile and Wireless Communications Technology, Afif Osseiran, Jose F. Monserrat, Patrick Marsch, 2nd Edition, Cambridge University Press, 2011
2. 5G NR: The Next Generation Wireless Access Technology, Erik Dahlman, Stefan Parkvall, Johan Skold, 1st Edition, Elsevier, 2016
3. Evolution of Air Interface for 5G, Suvra Sekhar Das, Ramjee Prasad, River, 2018

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC413) ELECTRONIC MEASUREMENTS AND INSTRUMENTATION

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE OBJECTIVES:

- To develop an awareness to various electronic measurement concepts
- To explain the operation and design of different signal generators
- To measurement using CRO's
- To measurement of electrical parameters using bridges
- To explain the operations using transducers

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the measurement and performance characteristics using electronic instruments

CO-2: Learn different signal generators, logic analyzers and harmonic distortion wave analyzers

CO-3: Measure amplitude and frequency using different types of CROs

CO-4: Measure resistance, capacitance and inductance using bridges

CO-5: Learn different types of transducers

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	3	2	2	3-	-	-	-	-	-	-	-	3	-	-
CO-2	3	3	2	2	3	-	-	-	-	-	-	-	3	-	-
CO-3	3	3	2	2	3	-	-	-	-	-	-	-	3	-	-
CO-4	3	3	2	2	3	-	-	-	-	-	-	-	3	-	-
CO-5	3	3	2	2	3	-	-	-	-	-	-	-	3	-	-

UNIT-I:

Performance characteristics of instruments, static characteristics, Accuracy, Resolution, Precision, Expected value, Errors, Sensitivity. Errors in measurement, Dynamic Characteristics, DC voltmeters. D'Arsonval Movement, DC Current meters, AC voltmeters and Current Meters, Range Extension. Design consideration of different types of Digital Voltmeters - Staircase Ramp- type DVM, Dual Slope integrating type DVM, Successive Approximation Type DVM.

UNIT-II:

Signal Generators: Fixed and variable, AF oscillators, Standard and AF sine and square wave signal generators, Function Generators, Square, pulse generator, Signal Analyzer, Logic Analyzer, Network Analyzer, Random noise generator, sweep generator, arbitrary waveform generator Wave analyzers, harmonic distortion wave analyzer, spectrum analyzers Frequency counter, time and period measurement.

UNIT-III:

Oscilloscopes: CRT Features, vertical amplifiers, Horizontal deflection system, sweep, trigger pulse, delay line, sync selector circuits, simple CRO, Trigger sweep CRO, dual beam CRO, Dual trace oscilloscope, sampling oscilloscope, storage oscilloscope, measurement of amplitude and frequency, probes for CRO-active and passive.

UNIT-IV:

DC Bridges: Wheatstone's bridge, Kelvin's bridge, AC bridges-measurement of inductance-Maxwell's bridge, Hay's bridge, measurement of capacitance, Schering bridge, Wien bridge, errors and precautions in using bridges, Q-meter, different connections in Q- meter.

UNIT-V:

Transducers: Active and passive transducers-Resistance transducers, Capacitance transducers, inductance transducers, Strain gauges transducers, LVDT transducers, Piezo electric transducers, Resistance thermometers, Thermocouples.

TEXT BOOKS:

1. Electronic Instrumentation, H. S. Kalsi, 2nd Edition, Tata McGraw-Hill, 2004
2. Modern Electronic Instrumentation and Measurement Techniques, A. D. Helfrick and W. D. Cooper, 5th Edition, Prentice Hall of India, 2002

REFERENCES:

1. Transducers and Display Systems, B. S. Sonde, 1977
2. Electronic Measurements and Instrumentation, B. M. Oliver and J.M. Cage, Tata McGraw-Hill, 2009
3. Electrical and Electronic Measurements, Shawney, Khanna Publications, 2016
4. Introduction to Instrumentation and Measurements, Robert Northrop, CRC Press, 2014
5. Electronic Measurements and Instrumentation, K. Lal Kishore, Pearson Education, 2005

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC414) LOW POWER VLSI SYSTEMS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: CMOS VLSI Design

COURSE OBJECTIVES:

- To study the concepts on different levels of power estimation and optimization techniques
- To learn various low-power design approaches
- To design the different low power combinational circuits
- To understand the design procedure of the low voltage and low power memories

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the different power dissipation sources in VLSI circuits

CO-2: Apply the low-power design approaches in voltage scaling and switched capacitance minimization

CO-3: Design energy-efficient adder circuits

CO-4: Design optimized multiplier circuits with low power dissipation

CO-5: Analyze the performance of low-power and low voltage memories

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	3	2	2	-	-	-	-	-	-	-	-	-	2	-	1
CO-2	3	2	2	-	-	-	-	-	-	-	-	-	2	-	2
CO-3	3	3	3	-	-	-	-	-	-	-	-	-	3	-	3
CO-4	3	3	3	-	-	-	-	-	-	-	-	-	3	-	3
CO-5	3	3	3	-	-	-	-	-	-	-	-	-	3	-	3

UNIT-I:

Fundamentals: Need for Low Power Circuit Design, Sources of Power Dissipation – Switching Power Dissipation, Short Circuit Power Dissipation, Leakage Power Dissipation, Glitching Power Dissipation

UNIT-II:

Short Channel Effects: Drain Induced Barrier Lowering and Punch Through, Surface Scattering, Velocity Saturation, Impact Ionization, Hot Electron Effect.

Low-Power Design Approaches: Low-Power Design through Voltage Scaling – VTCMOS circuits, MTCMOS circuits, Architectural Level Approach –Pipelining and Parallel Processing Approaches. Switched Capacitance Minimization Approaches

UNIT-III:

Low-Voltage Low-Power Adders: Introduction, Standard Adder Cells, CMOS Adder's Architectures – Ripple Carry Adders, Carry Look- Ahead Adder, Carry Select Adders, Carry Save Adders

UNIT-IV:

Low-Voltage Low-Power Multipliers: Introduction, Overview of Multiplication, Types of Multiplier Architectures, Braun Multiplier, Baugh- Wooley Multiplier, Booth Multiplier.

UNIT-V:

Low-Voltage Low-Power Memories: Basics of ROM, Low-Power ROM Technology, Future Trend and Development of ROMs, Basics of SRAM, Memory Cell, Precharge and Equalization Circuit, Low-Power SRAM Technologies, Basics of DRAM, Self-Refresh Circuit, Future Trend and Development of DRAM.

TEXT BOOKS:

1. CMOS Digital Integrated Circuits – Analysis and Design, Sung-Mo Kang, Yusuf Leblebici, Tata McGraw-Hill, 2011
2. Low-Voltage, Low-Power VLSI Subsystems, Kiat-Seng Yeo, Kaushik Roy, McGraw-Hill

REFERENCES:

1. Introduction to VLSI Systems: A Logic, Circuit and System Perspective, Ming-BO Lin, CRC Press, 2011
2. Low Power CMOS Design, Anantha Chandrakasan, IEEE Press/Wiley International, 1998
3. Low Power CMOS VLSI Circuit Design, Kaushik Roy, Sharat C. Prasad, John Wiley & Sons, 2000

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. VIII Semester

(22PE1EC415) IOT AND RASPBERRY PI PROGRAMMING

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	5	5	60	100

COURSE PRE-REQUISITES: Microprocessor and Microcontrollers, Python Programming and Practice

COURSE OBJECTIVES:

- To understand the Raspberry Pi platform
- To acquire a foundation in tailor made Python for Raspberry Pi
- To understand the low-level hardware interaction and system-level programming
- To learn and implement Internet of Things (IoT) solutions using Raspberry Pi

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand the hardware and software requirements of Raspberry Pi

CO-2: Program Raspberry Pi using python

CO-3: Explore on board components of Raspberry Pi

CO-4: Interface various sensors and communication modules with Raspberry Pi

CO-5: Build end-to-end IoT systems on the Raspberry Pi

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)												PROGRAM SPECIFIC OUTCOMES (PSO)		
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PO-12	PSO-1	PSO-2	PSO-3
CO-1	1	3	2	-	-	-	-	-	-	-	-	-	1	3	3
CO-2	2	3	2	-	-	-	-	-	-	-	-	-	3	3	3
CO-3	2	3	3	-	2	-	-	-	-	-	-	-	3	3	3
CO-4	2	3	3	-	2	-	-	-	-	-	-	-	3	3	3
CO-5	2	3	3	3	2	2	2	1	-	-	-	2	3	3	3

UNIT-I:

Overview of Raspberry Pi: Introduction to Raspberry Pi, Comparison of various Raspberry Pi Models, Understanding SoC architecture and SoCs used in Raspberry Pi, Pin Description and On-board components of Raspberry Pi, Booting up Raspberry Pi - operating System and Linux Commands, Linux- Introduction, Architecture, File System.

Raspbian Operating System: Introduction, Tools like Leaf pad Editor, Installing Raspbian on Pi, First boot and Basic Configuration of Pi, Popular Linux Commands.

UNIT-II:

Raspberry Pi Programming Using Python: Introduction, Python versus other languages, applications of Python, keywords, operators and operands, data types in Python, importing libraries, flow control, Conditional Statement, Loops.

UNIT-III:

The GPIO Port: GPIO Pinout Diagrams, GPIO Features, UART Serial Bus, I2C Bus, SPI Bus, Using the GPIO Port in Python, GPIO Output - Flashing an LED, GPIO Input - Reading a Button.

The Raspberry Pi Camera Module: Introduction to the Raspberry Pi Camera Module, installation, application to photography and video recording.

UNIT-IV:

Sensors Interfacing: Temperature and Humidity Sensor (DHT11), Motion Sensor (PIR), Obstacle detection using Ultrasonic sensor.

Communicating using Raspberry Pi: GSM interfacing, accessing on-board Wi-Fi, Connecting Database with Raspberry Pi.

UNIT-V:

IoT Design using Raspberry Pi: IoT Applications based on Pi, Home automation using Pi, Node-RED, MQTT Protocol, Using Node-RED Visual Editor on Raspberry Pi

TEXT BOOKS:

1. Programming the Raspberry Pi: Getting Started with Python, Simon Monk, 2nd Edition, McGraw-Hill Professional
2. Exploring Raspberry Pi: Interfacing to the Real World with Embedded Linux, 1st Edition, Derek Molloy, Wiley, 2016

REFERENCES:

1. The Computer Programming Bible, Cyberpunk Architects, 2020
2. Raspberry Pi IoT Projects: Prototyping Experiments for Makers, John C. Shovic, APress, 2016